Academic Statement of Purpose

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Note: I legally changed my name from Sangwoo Cho to Sangwoo Shin, but some of my cited publications may appear under my previous name.

Research Interests

My primary research interest is language grounding for embodied control, particularly in integrating large language models (LLMs) into robots to perform complex tasks in diverse and dynamic environments. LLMs, with their vast internet-scale knowledge, hold great promise for this, but two key challenges must be addressed: (i) LLM's inherent lack of environmental knowledge and (ii) the need for fine-tuning control systems in response to specific environmental conditions.

Thus, during my Ph.D., I aim to develop an efficient, unified system that harmonizes LLMs and low-level controllers, enabling robots to seamlessly translate high-level language instructions into precise, context-aware actions.

Grounding LLM Reasoning in Physical Environments

The process of connecting LLM's commonsense reasoning capabilities to real-world environments—what we call grounding—is crucial for its applications. One promising direction in this field is leveraging LLM's ability to decompose complex tasks into controller-executable subtasks, a process known as embodied task planning. Yet, as LLMs are pretrained primarily on static text data, their planning decisions often lack the environmental awareness necessary for operating across diverse environments.

In response to this challenge, my colleagues and I investigated the cross-domain adaptation of LLM agents in embodied environments [1]. We identified a fundamental trade-off between the semantic richness and domain-agnostic executability of robotic skills. For instance, high-level semantic skills like "prepare a meal" encompass richer semantic content and longer-horizon behaviors compared to low-level semantic skills such as "grab a cup". However, these high-level skills tend to be effective only within specific environment domains, as their execution heavily depends on numerous environmental factors, unlike their low-level counterparts which can be universally executable across domains. Based on these insights, we developed an iterative embodied task planning algorithm that starts with higher levels of semantic skill hierarchy and then progressively moves downwards, so as to actualize each planned skill into physical actions within the target environment.

In [2], we leveraged LLMs as reward estimators for offline learning of parameter-efficient embodied agents. Here, to ensure that LLM-generated rewards are grounded in specific environmental contexts, we introduced an adaptive ensemble of spatio-temporally consistent rewards, followed by an alignment process with labeled sparse rewards. The resulting agent trained by offline RL is capable of performing complex tasks with high efficiency and minimal latency in diverse environments, overcoming the limitations faced by the agents that rely on the online exploitation of LLMs.

Building upon these experiences, a key direction that I want to explore is to develop embodied LLM agents that can dynamically adapt to non-stationary environments where the environmental conditions constantly change, often beyond the agent's immediate perception. To address this problem, I plan to investigate a RAG (Retrieval-Augmented Generation) system tailored for embodied reasoning. While most NLP tasks focus on sourcing and utilizing information from static databases, embodied tasks require a continuously updated database that integrates only the most relevant environmental information. My goal is to design a system that dynamically determines which data to save and retrieve, how frequently it should be updated, and in what format (e.g., text, knowledge

graph) it can best support LLM-driven embodied task planning.

Embodied Control with Foundation Models

The advancement of foundation models has opened new branches of embodied control. However, a significant challenge remains in learning robot skills that can effectively translate LLM-planned strategies into physical actions.

In [3], we proposed a method for efficiently extracting robot skills from offline datasets and learning them by leveraging the pretrained Vision Language Models (VLMs). To facilitate the adaptation capability of these skills to various domain diversities inherent in non-stationary environments, we also developed an unsupervised meta-learning technique for encoding environmental dynamics. Our work in [4] further explored the concept of the robot skill's zero-shot adaptation in cross-domain settings, where user input in interleaved multi-modal snippets can prompt complex tasks for different domains. In response, we proposed a hierarchical adaptation technique, leveraging the modality transfer capability of VLMs in conjunction with the reasoning prowess of LLMs.

To ground LLM's wealth of actionable knowledge and enable LLM to physically act in the real world, the community has focused on two primary directions. The first is the exploration of a large-scale Vision-Language-Action (VLA) model by augmenting pretrained VLMs with action tokens. Building upon my prior distillation work [5], I plan to explore efficient inference techniques (e.g., distillation, pruning, RAG) and reasoning strategies (e.g., Chain-of-Thought) for VLAs. Additionally, I am interested in efficiently leveraging crowdsourced, diverse-quality datasets for the training of VLA models [6].

Another approach leverages LLM's code generation ability to invoke low-level controller APIs rather than directly generating fine-grained action trajectories. Expanding on this, I aim to develop a more principled framework by introducing systemic abstractions between LLMs and controller APIs. This framework will decouple LLM's high-level contextual understanding from the external control tool's reliability, enabling more rigorous verification and control of action execution. I believe this decoupling strategy would enable the development of more robust and adaptable agents that can operate across open domains without additional training in the foundation model.

Ph.D. Studies at UW Madison and Career Goals

I believe the UW Madison CS program is an ideal fit for my research aspirations. Specifically, I am drawn to Professor **X**'s work on foundation models. I plan to explore how his expertise in efficient vision-language models can enhance embodied agents' decision-making capabilities. I am also excited about the possibility of collaborating with **Y** on developing a robust RAG system for embodied reasoning and advancing tool-use approaches for LLM-guided embodied control. Lastly, I highly value the collaborative and inclusive environment of the UW Madison CS program, which I believe will help me expand my research perspective and conduct highly impactful research. After a Ph.D., I plan to pursue an academic career, since I believe universities offer more creative freedom and open-ended research opportunities.

References

[1] **Sangwoo Shin***, Seunghyun Kim*, Youngsoo Jang, Moontae Lee, Honguk Woo. Semantic Skill Grounding for Embodied Instruction-Following in Cross-Domain Environments. ACL 2024 Findings.

[2] Yujeong Lee*, **Sangwoo Shin***, Wei-Jin Park, Honguk Woo. LLM-Based Offline Learning for Embodied Agents via Consistency-Guided Reward Ensemble. EMNLP 2024 Findings.

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[6] Minjong Yoo, **Sangwoo Cho**, Honguk woo. Skills Regularized Task Decomposition for Multi-task Offline Reinforcement Learning. Neurips 2022.